Field	Entry
Source ID	WBC-0014
Title	A Study on Monitoring of Frying Performance and Oxidative Stability of Virgin Coconut Oil (VCO) During Continuous/Prolonged Deep Fat Frying Process Using Chemical and FTIR Spectroscopy
Source type	Paper
Link / File	DOI: 10.1007/s13197-013-1078-8
Craft applicability	Wood (surface-treatment oil used on food-contact Udayagiri products)
Primary Why-theme	11. Product Function, Use and Performance
Process stage	Final treatment (oil application)
Questions answered / generated	Why can't coconut oil's food-safe status alone guarantee it is a stable long-term wood finish? Why might oil-treated food-contact objects need regular reapplication or maintenance?
Evidence level	E3 (direct scientific evidence, chemical/FTIR measurement of oxidative degradation)
Relevance note	Directly measures that virgin coconut oil's free fatty acid content and peroxide value increase significantly with prolonged heat/time exposure, evidencing progressive oxidative degradation — supporting the report's caution that coconut oil's performance as a wood finish on food-contact Udayagiri products (repeated washing, storage, use) requires separate testing rather than assumption from its edibility.